MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

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Review Questions Part 4

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These are review questions for Chapter 7 and Chapter 8 in Ross' book.

Tutorial Questions

(1) Suppose that X and Y are both Bernoulli random variables. Show that X and Y are independent if and only if $\operatorname{Cov}(X,Y)=0$.

Solution: We already know that $\operatorname{Cov}(X,Y)=0$ if X and Y are independent. Now suppose that $\operatorname{Cov}(X,Y)=0$, then by

$$\operatorname{Cov}(X,Y)=\mathbb{E}(XY)-\mathbb{E}(X)\,\mathbb{E}(Y),$$

we have $\mathbb{E}(XY)=\mathbb{E}(X)\,\mathbb{E}(Y)$, and hence, by the assumption that X and Y are both Bernoulli random variables, we have $\mathbb{E}(XY)=P\{X=1,Y=1\}$, $\mathbb{E}(X)=P\{X=1\}$, $\mathbb{E}(Y)=P\{Y=1\}$, and thus $P\{X=1,Y=1\}=P\{X=1\}P\{Y=1\}$.

By $\operatorname{Cov}(X,Y)=0$, we have $\operatorname{Cov}(1-X,Y)=\operatorname{Cov}(X,1-Y)=\operatorname{Cov}(1-X,1-Y)=0$, e.g.,

- $\operatorname{Cov}(1-X,Y)=\mathbb{E}((1-X)Y)-\mathbb{E}(1-X)\,\mathbb{E}(Y)=\mathbb{E}(1Y)-\mathbb{E}(XY)-(1-\mathbb{E}(X))\,\mathbb{E}(Y)$ $=\mathbb{E}(Y)-\mathbb{E}(XY)-\mathbb{E}(Y)+\mathbb{E}(X)\,\mathbb{E}(Y)=-\operatorname{Cov}(X,Y)=0.$

This gives

- $P\{X=0,Y=1\}=P\{X=0\}P\{Y=1\}$.

Similarly, by $\operatorname{Cov}(X,1-Y)=\operatorname{Cov}(1-X,1-Y)=0$, we have $P\{X=1,Y=0\}=P\{X=1\}P\{Y=0\}$, $P\{X=0,Y=0\}=P\{X=0\}P\{Y=0\}$.

This shows that X and Y are independent.

(2) The nine players on a basketball team consist of 2 centers, 3 forwards, and 4 backcourt players. If the players are paired up at random into three groups of size 3 each, find
(a) the expected value and (b) the variance of the number of triplets consisting of one of each type of player.

Solution: Let X_i equal 1 if the i-th triple consists of one of each type of player. Then

$$\mathbb{E}(X_i)=\frac{\binom{2}{1}\binom{3}{1}\binom{4}{1}}{\binom{9}{3}}=\frac{2}{7}.$$

Hence, for part (a), we obtain $\mathbb{E}\left(\sum_{i=1}^{3}X_i\right)=\sum_{i=1}^{3}\mathbb{E}(X_i)=\frac{6}{7}$.

We also have $\mathrm{Var}(X_i) = \left(\frac{2}{7}\right)\left(1 - \frac{2}{7}\right) = \frac{10}{49}$. To calculate $\mathrm{Var}\left(\sum_{i=1}^{3} X_i\right)$, we also need to have $\mathrm{Cov}(X_i, X_j)$ for $i \neq j$. We calculate $\mathrm{Cov}(X_1, X_2)$ by

$$\begin{aligned}
\mathrm{Cov}(X_1, X_2) &= \mathbb{E}(X_1 X_2) - \mathbb{E}(X_1)\,\mathbb{E}(X_2) \\
&= P\{X_1 = 1, X_2 = 1\} - P\{X_1 = 1\}\,\mathbb{E}\{X_2 = 1\} \\
&= P\{X_1 = 1\}P\{X_2 = 1 \mid X_1 = 1\} - \frac{2}{7} \times \frac{2}{7} \\
&= \frac{2}{7} \times \frac{\binom{1}{1}\binom{2}{1}\binom{3}{1}}{\binom{6}{3}} - \frac{4}{49} \\
&= \frac{6}{70} - \frac{4}{49} \\
&= \frac{1}{245}.
\end{aligned}$$

We can have for any $i, j \in \{1, 2, 3\}$ with $i \neq j$, $\mathrm{Cov}(X_i, X_j) = \frac{1}{245}$. Thus, $\mathrm{Var}\left(\sum_{i=1}^{3} X_i\right) = \sum_{i=1}^{3} \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j) = \frac{30}{49} + \frac{6}{245} = \frac{156}{245}$.

(3) If a die is to be rolled until all sides have appeared at least once, find the expected number of times that outcome 1 appears.

Solution: It is related to the problem we introduced in lecture, the coupon collector problem. Here we have six outcomes for each rolling, and we know that the expected number of times that the die need be rolled until all sides have appeared at least once is

$$6(1 + 1/2 + 1/3 + 1/4 + 1/5 + 1/6) = 14.7.$$

Now, if we let X_i denote the total number of times that side i appears, then, since $\sum_{i=1}^{6} X_i$ is equal to the total number of rolls, we have

$$14.7 = \mathbb{E}\left(\sum_{i=1}^{6} X_i\right) = \sum_{i=1}^{6} \mathbb{E}(X_i).$$

By symmetry, $\mathbb{E}(X_i)$ will be the same for all i, and thus it follows from the preceding that $\mathbb{E}(X_i) = 14.7/6 = 2.45$.

(4) A deck of n cards numbered 1 through n, initially in any arbitrary order, is shuffled in the following manner: At each stage, we randomly choose one of the cards and move it to the front of the deck, leaving the relative positions of the other cards unchanged. This procedure is continued until all but one of the cards has been chosen. At this point it follows by symmetry that all $n!$ possible orderings are equally likely. Find the expected number of stages that are required.

Solution: This is a problem related to the Coupon Collector Problem introduced in the lecture, and what we want to get is just the expected time to collect $n - 1$ of the n types of coupons in the Coupon Collector Problem. By the results of that example, the solution is

$$1 + \frac{n}{n-1} + \frac{n}{n-2} + \cdots + \frac{n}{2}.$$

(5) A deck of 52 cards is shuffled and a bridge hand of 13 cards is dealt out. Let X and Y denote, respectively, the number of aces and the number of spades in the hand.

(a) Show that X and Y are uncorrelated.

(b) Are they independent?

Solution: For any $i, j \in \{1, 2, \cdots, 13\}$, let X_i be equal to 1 if the i-th card is an ace and let it be 0 otherwise, and let Y_j be equal to 1 if the j-th card is a spade and let it be 0 otherwise.

For any i, j, X_i is clearly independent of Y_j because knowing the suit of a particular card gives no information about whether it is an ace and thus cannot affect the probability that another specified card is an ace.

Thus, $\mathrm{Cov}(X_i, Y_j) = 0$, for any i, j, which gives

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}\left(\sum_{i=1}^{13} X_i, \sum_{j=1}^{13} Y_j\right) = \sum_{i=1}^{13} X_i \sum_{j=1}^{13} \mathrm{Cov}(X_i, Y_j) = 0.$$

X and Y are not independent, by

$$P\{Y = 13 \mid X = 4\} = 0 \neq P\{Y = 13\},$$

and hence

$$P\{Y = 13, X = 4\} \neq P\{Y = 13\}P\{X = 4\}.$$

(6) Let X be the length of the initial run in a random ordering of n ones and m zeroes. That is, if the first k values are the same (either all ones or all zeroes), then $X \geq k$. Find $\mathbb{E}(X)$.

Solution: Let L be the length of the initial run. Conditioning on the first value gives

$$\mathbb{E}(L) = \mathbb{E}(L \mid \text{first value is } 1)\frac{n}{n+m} + \mathbb{E}(L \mid \text{first value is } 0)\frac{m}{n+m}.$$

- If the first value is one, then the length of the run will be the position of the first zero when considering the remaining $n + m - 1$ values, of which $n - 1$ are ones and m are zeroes.
  $\mathbb{E}(L \mid \text{first value is } 1)$ is calculated as follows: We can number the $n-1$ ones left as $1_1, 1_2, \cdots, 1_{n-1}$, and for $i = 1, \cdots, n-1$, let A_i denote the event that 1_i appears before the the first 0 appears. Then $X = 1 + A_1 + \cdots + A_{n-1}$ (1 for the first value in this run), which is equal to $1 + \sum_{i=1}^{n-1} P\{A_i = 1\}$. Now for i, we consider 1_i and m many zeros, and $P\{A_i = 1\}$ is the probability that 1_i appears before all m many zeros, which is equal to $\frac{1}{m+1}$. This gives $\mathbb{E}(L \mid \text{first value is } 1) = 1 + \sum_{i=1}^{n-1} P\{A_i = 1\} = 1 + \frac{n-1}{m+1} = \frac{m+n}{m+1}$.
- If the first value is zero, then the length of the run will be the position of the first one when considering the remaining $n+m-1$ values, of which n are ones and $m-1$ are zeroes. By the same argument above, we have $\mathbb{E}(L \mid \text{first value is } 0) = \frac{m+n}{n+1}$.

Thus,

$$\begin{aligned}\mathbb{E}(L) &= \mathbb{E}(L \mid \text{first value is } 1)\frac{n}{n+m} + \mathbb{E}(L \mid \text{first value is } 0)\frac{m}{n+m} \\ &= \frac{m+n}{m+1} \times \frac{n}{n+m} + \frac{m+n}{n+1} \times \frac{m}{n+m} = \frac{n}{m+1} + \frac{m}{n+1}.\end{aligned}$$

(7) A deck of $2n$ cards consists of n red and n black cards. The cards are shuffled and then turned over one at a time. Suppose that each time a red card is turned over, we win 1 unit if more red cards than black cards have been turned over by that time. (For instance, if $n = 2$ and the result is $rbrb$, then we would win a total of 2 units.) Find the expected amount that we win.

Solution: Let I_j be equal to 1 if we win 1 when the j-th red card to show is turned over, and let it be equal to 0 otherwise. Hence, if X is our total winnings, then

$$\mathbb{E}(X) = \mathbb{E}\left(\sum_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} \mathbb{E}(I_j).$$

Now, I_j will equal 1 if j red cards appear before j black cards. By symmetry, the probability of this event is equal to $1/2$; therefore, $\mathbb{E}(I_j) = 1/2$ and $\mathbb{E}(X) = n/2$.

(8) Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.
(a) What can be said about the probability that this week's production will exceed 75?
(b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution: Let X be the number of items that will be produced in a week.
(a) By Markov's inequality,

$$P\{X > 75\} = P\{X \geq 76\} \leq \frac{\mathbb{E}(X)}{76} = \frac{50}{76} = \frac{25}{38}.$$

(b) By Chebyshev's inequality,

$$P\{|X - 50| \geq 10\} \leq \frac{\sigma^2}{10^2} = \frac{25}{100} = \frac{1}{4}.$$

This gives

$$P\{40 < X < 60\} = P\{|X - 50| < 10\} = 1 - P\{|X - 50| \geq 10\} \geq 1 - \frac{1}{4} = \frac{3}{4}.$$

Thus, the probability that this week's production will be between 40 and 60 is at least .75.

(9) Suppose that the number of units produced daily at factory A is a random variable with mean 20 and standard deviation 3 and the number produced at factory B is a random variable with mean 18 and standard deviation 6. Assuming independence, derive an upper bound for the probability that more units are produced today at factory B than at factory A.

Solution: Let X be the number produced at factory A and Y be the number produced at factory B, then

$$\mathbb{E}(Y - X) = -2, \qquad \mathrm{Var}(Y - X) = \mathrm{Var}(Y) + \mathrm{Var}(X) = 36 + 9 = 45.$$

This gives

$$P\{Y - X > 0\} = P\{Y - X \geq 1\} = P\{Y - X + 2 \geq 3\} \leq P\{|Y - X + 2| \geq 3\} \leq \frac{\mathrm{Var}(Y - X)}{3^2} = 5,$$

which is a trivial bound.

The following is optional: for this question, as $\mathbb{E}(Y - X + 2)$ has mean 0, we can apply the one-sided Chebyshev's inequality to obtain

$$P\{Y - X > 0\} \leq \frac{\mathrm{Var}(Y - X + 2)}{\mathrm{Var}(Y - X + 2) + 3^2} = \frac{45}{45 + 9} = \frac{5}{6}.$$

(10) An astronomer is interested in measuring the distance, in light-years, from his observatory to a distant star. Although the astronomer has a measuring technique, he knows that, because of changing atmospheric conditions and normal error, each time a measurement is made it will not yield the exact distance, but merely an estimate. As a result, the astronomer plans to make a series of measurements and then use the average value of these measurements as his estimated value of the actual distance. If the astronomer believes that the values of the measurements are independent and identically distributed random variables having a common mean d (the actual distance) and a common variance of 4 (light-years), how many measurements need he make to be reasonably sure that his estimated distance is accurate to within $\pm.5$ lightyear?

Solution: Here we consider "reasonably sure" as "95% sure".

Suppose that the astronomer decides to make n observations. If $X_1, X_2, \cdots, X_n$ are the n measurements, then, from the central limit theorem, it follows that

$$Z_n \sim \frac{\sum_{i=1}^{n} X_i - nd}{2\sqrt{n}}$$

has approximately a standard normal distribution. Hence

$$P\left\{-0.5 \leq \frac{\sum_{i=1}^{n} X_i}{n} - d \leq 0.5\right\} = P\left\{-0.5 \times \frac{\sqrt{n}}{2} \leq Z_n \leq 0.5 \times \frac{\sqrt{n}}{2}\right\}$$

$$\approx \Phi\left(\frac{\sqrt{n}}{4}\right) - \Phi\left(-\frac{\sqrt{n}}{4}\right) = 2\Phi\left(\frac{\sqrt{n}}{4}\right) - 1.$$

Therefore, if the astronomer wants, for instance, to be 95 percent certain that his estimated value is accurate to within 0.5 light year, he should make n^* measurements, where n^* is such that

$$2\Phi\left(\frac{\sqrt{n^*}}{4}\right) - 1 = 0.95, \qquad \Phi\left(\frac{\sqrt{n^*}}{4}\right) = 0.975.$$

This gives $\frac{\sqrt{n^*}}{4} = 1.96$ and $n^* = (7.84)^2 \approx 61.47$. As n^* is not integral valued, he should make 62 observations.